



DUBLIN CITY UNIVERSITY

SEMESTER 1 EXAMINATIONS 2017/2018

MODULE: CA208 - Logic

PROGRAMME(S):

CASE BSc in Computer Applications (Sft.Eng.)
CPSSD BSc in Computational Problem Solving and SW Dev.

YEAR OF STUDY: 2

EXAMINERS: David Sinclair (Ext:5510)

TIME ALLOWED: 2 hours

INSTRUCTIONS: Answer ALL 6 questions from Section A. Answer 2 out of 3 questions from Section B. Question 1 carries 30 marks. Questions 2 to 9 carry 10 marks.

PLEASE DO NOT TURN OVER THIS PAGE UNTIL INSTRUCTED TO DO SO

The use of programmable or text storing calculators is expressly forbidden.
Please note that where a candidate answers more than the required number of questions, the examiner will mark all questions attempted and then select the highest scoring ones.

There are no additional requirements for this paper.

Section A

QUESTION 1

[Total marks: 30]

Every child loves Santa. Everyone who loves Santa loves reindeer. Rudolph is a reindeer and Rudolph has a red nose. Anything that has a red nose is weird or a clown. No reindeer is a clown. Scrooge does not loves anything which is weird.

1(a) [10 Marks]

Convert the statements above into a set of predicate formulae.

1(b) [10 Marks]

Convert the predicate formulae from Question 1(a) into *Prenex Conjunctive Normal Form (PCNF)*.

1(c) [10 Marks]

Using *General Resolution* prove that Scrooge is not a child.

[End Question 1]

QUESTION 2

[Total marks: 10]

[10 Marks]

Using semantic tableaux, prove the following propositional formula is valid.

$$(p \rightarrow q) \rightarrow ((p \rightarrow \neg q) \rightarrow \neg p)$$

[End Question 2]

QUESTION 3

[Total marks: 10]

[10 Marks]

Using *semantic tableaux* prove the following proposition is valid.

$$\neg(p \wedge q) \leftrightarrow \neg p \vee \neg q$$

[End Question 3]

QUESTION 4**[Total marks: 10]**

[10 Marks]

Using the Gentzen System $\mathcal{G}$ prove

$$\vdash (A \rightarrow B) \rightarrow ((\neg A \rightarrow B) \rightarrow B)$$

[End Question 4]**QUESTION 5****[Total marks: 10]**

[10 Marks]

Using the Gentzen System $\mathcal{G}$ prove

$$\exists x.(p(x) \rightarrow q(x)) \rightarrow (\forall x.p(x) \rightarrow \exists x.q(x))$$

[End Question 5]**QUESTION 6****[Total marks: 10]**

[10 Marks]

Convert the following propositional formula into *conjunctive normal form (CNF)*. Clearly identify each step in the process.

$$((p \leftrightarrow q) \rightarrow (r \rightarrow s))$$

[End Question 6]

Section B

QUESTION 7

[Total marks: 10]

[10 Marks]

Given the following Prolog relations:

- `parent(X,Y)` is true if X is the parent of Y,
- `male(X)` is true if X is male, and
- `female(X)` is true if X is female,

write the following Prolog relations:

- `uncle(X,Y)` is true if X is an uncle of Y,
- `greatuncle(X,Y)` is true if X is the uncle of a parent of Y,
- `ancestor(X,Y)` is true if X is a direct ancestor of Y.

[End Question 7]

QUESTION 8

[Total marks: 10]

[10 Marks]

Without using any built-in string predicates, write the following predicates:

- `concat(A, B, C)` is true if string C is the concatenation of strings A and B.
- `reverse(A, B)` is true if string B is the reverse of string A.

[End Question 8]

QUESTION 9

[Total marks: 10]

[10 Marks]

Describe the procedural interpretation of a Prolog program.

[End Question 9]

[APPENDICES]

Semantic Tableaux

α	α_1	α_2	β	β_1	β_2
$\neg\neg A_1$	A_1		$\neg(B_1 \wedge B_2)$	$\neg B_1$	$\neg B_2$
$A_1 \wedge A_2$	A_1	A_2	$B_1 \vee B_2$	B_1	B_2
$\neg(A_1 \vee A_2)$	$\neg A_1$	$\neg A_2$	$B_1 \rightarrow B_2$	$\neg B_1$	B_2
$\neg(A_1 \rightarrow A_2)$	A_1	$\neg A_2$	$B_1 \uparrow B_2$	$\neg B_1$	$\neg B_2$
$\neg(A_1 \uparrow A_2)$	A_1	A_2	$\neg(B_1 \downarrow B_2)$	B_1	B_2
$A_1 \downarrow A_2$	$\neg A_1$	$\neg A_2$	$\neg(B_1 \leftrightarrow B_2)$	$\neg(B_1 \rightarrow B_2)$	$\neg(B_2 \rightarrow B_1)$
$A_1 \leftrightarrow A_2$	$A_1 \rightarrow A_2$	$A_2 \rightarrow A_1$	$B_1 \oplus B_2$	$\neg(B_1 \rightarrow B_2)$	$\neg(B_2 \rightarrow B_1)$
$\neg(A_1 \oplus A_2)$	$A_1 \rightarrow A_2$	$A_2 \rightarrow A_1$			

γ	$\gamma(a)$
$\forall x.A(x)$	$A(a)$
$\neg\exists x.A(x)$	$\neg A(a)$

δ	$\delta(a)$
$\exists x.A(x)$	$A(a)$
$\neg\forall x.A(x)$	$\neg A(a)$

Gentzen Logic

$$\frac{\vdash U'_1 \cup \{\alpha_1, \alpha_2\}}{\vdash U'_1 \cup \{\alpha\}} \quad \frac{\vdash U'_1 \cup \{\beta_1\} \quad \vdash U'_2 \cup \{\beta_2\}}{\vdash U'_1 \cup U'_2 \cup \{\beta\}}$$

$$\frac{\vdash U'_1 \cup \{\gamma, \gamma(a)\}}{\vdash U'_1 \cup \{\gamma\}} \quad \frac{\vdash U'_1 \cup \{\delta(a)\}}{\vdash U'_1 \cup \{\delta\}}$$

$\frac{A}{B}$ is a *sequent* and means that if A is true, then B can be inferred.

α	α_1	α_2	β	β_1	β_2
$\neg\neg A$	A		$B_1 \wedge B_2$	B_1	B_2
$\neg(A_1 \wedge A_2)$	$\neg A_1$	$\neg A_2$	$\neg(B_1 \vee B_2)$	$\neg B_1$	$\neg B_2$
$A_1 \vee A_2$	A_1	A_2	$\neg(B_1 \rightarrow B_2)$	B_1	$\neg B_2$
$A_1 \rightarrow A_2$	$\neg A_1$	A_2	$\neg(B_1 \uparrow B_2)$	B_1	B_2
$A_1 \uparrow A_2$	$\neg A_1$	$\neg A_2$	$B_1 \downarrow B_2$	$\neg B_1$	$\neg B_2$
$\neg(A_1 \downarrow A_2)$	A_1	A_2	$B_1 \leftrightarrow B_2$	$B_1 \rightarrow B_2$	$B_2 \rightarrow B_1$
$\neg(A_1 \leftrightarrow A_2)$	$\neg(A_1 \rightarrow A_2)$	$\neg(A_2 \rightarrow A_1)$	$\neg(B_1 \oplus B_2)$	$B_1 \rightarrow B_2$	$B_2 \rightarrow B_1$
$A_1 \oplus A_2$	$\neg(A_1 \rightarrow A_2)$	$\neg(A_2 \rightarrow A_1)$			

γ	$\gamma(a)$
$\exists x.A(x)$	$A(a)$
$\neg\forall x.A(x)$	$\neg A(a)$

δ	$\delta(a)$
$\forall x.A(x)$	$A(a)$
$\neg\exists x.A(x)$	$\neg A(a)$

[END OF APPENDICES]

[END OF EXAM]